EXP-17 : To generate DES subkeys directly in decryption order K16 to K1.

PROGRAM :

Run | Debug | Stop | Share | Save | { } Beautify

main.c

```c
#include <stdio.h>
#include <stdint.h>
int pc1[]={57,49,41,33,25,17,9,1,58,50,42,34,26,18,
 10,2,59,51,43,35,27,19,11,3,60,52,44,36,63,55,
 47,39,31,23,15,7,62,54,46,38,30,22,14,6,61,53,
 45,37,29,21,13,5,28,20,12,4};
int pc2[]={14,17,11,24,1,5,3,28,15,6,21,10,23,19,12,4,
 26,8,16,7,27,20,13,2,41,52,31,37,47,55,30,40,
 51,45,33,48,44,49,39,56,34,53,46,42,50,36,29,32};
uint64_t perm(uint64_t x,int *p,int n,int bits) {
 uint64_t y=0;
 for (int i=0;i<n;i++) y=(y<<1)|((x>>(bits-p[i]))&1);
 return y;
}
uint32_t right(uint32_t x,int s) {
 return ((x>>s)|(x<<(28-s)))&0x0FFFFFFF;
}
int main(void) {
 int shift[]={1,1,2,2,2,2,2,2,1,2,2,2,2,2,2,1};
 uint64_t k=perm(0x133457799BBCDFF1ULL,pc1,56,64);
 uint32_t c=(uint32_t)(k>>28), d=k&0x0FFFFFFF;
 for (int i=15;i>=0;i--) {
 uint64_t sub=perm(((uint64_t)c<<28)|d,pc2,48,56);
 printf("K%02d=%012llX%s",i+1,
 (unsigned long long)sub,i%2==0 ? "\n":" ");
 c=right(c,shift[i]); d=right(d,shift[i]);
 }
 return 0;
}
```

OUTPUT :

```
K14=5F43B7F2E73A K13=97C5D1FABA41
K12=7571F59467E9 K11=215FD3DED386
K10=B1F347BA464F K09=E0DBEBEDE781
K08=F78A3AC13BFB K07=EC84B7F618BC
K06=63A53E507B2F K05=7CEC07EB53A8
K04=72ADD6DB351D K03=55FC8A42CF99
K02=79AED9DBC9E5 K01=1B02EFFC7072


...Program finished with exit code 0
Press ENTER to exit console.
```